CEREBRALZIP --- INTERNSHIP ASSIGNMENT

Forensic License Plate Enhancement, Restoration & OCR

Technical Approach, Quantitative Evaluation and Failure Analysis
Prepared for submission --- November 2026

Evaluation Scope

Item Value

Real-world evaluation set 300 images

Supplementary plate-only 18 reported samples image-quality benchmark

Execution device recorded by `cuda:0` evaluator

Mean end-to-end latency 3,713 ms/image

Throughput 0.27 FPS

1. Executive Summary

This project implements a forensic-oriented automatic number plate recognition (ANPR) pipeline. For each image, the system aims to localize the license plate, produce a corrected/enhanced plate crop, and recover the registration text. The image pipeline combines full-scene motion deblurring, YOLOv8 plate detection, geometric rectification, image-quality assessment, adaptive enhancement candidates, dual-polarity OCR, and syntax-aware candidate selection. A separate video mode adds vehicle tracking and observation deduplication.

The primary quantitative evaluation was performed on 300 real-world images. The detector achieved **99.0% recall at IoU 0.50** and **AP@0.50 of 0.9873**. End-to-end exact-match OCR accuracy was **50.0%**, with corpus CER of **0.1811**. Mean end-to-end latency was **3.713 seconds per image (0.27 FPS)**. Thus, localization is strong, while exact text recovery and runtime remain the principal system-level limitations.

The available 300-image set was predominantly clean or only somewhat degraded; it did not contain a sufficiently large, controlled population of severely blurred/noisy plates. Consequently, aggregate OCR results on this set do not fully characterize enhancement behaviour under severe degradation. During additional qualitative testing, the pipeline was observed to make some very poor-quality plates more readable and, in selected cases, support recovery of text that was difficult to read in the original. These observations are

reported as **qualitative stress-test findings**, not as a statistically established improvement on the 300-image benchmark.

A supplementary plate-only image-quality benchmark contains 18 paired records. As provided, mean PSNR increased from **13.09 dB to 18.38 dB (+5.28 dB)**, and mean SSIM increased from **0.3695 to 0.7458 (+0.3763)**. These figures should be interpreted as restoration-fidelity evidence only if the reference images used for both measurements are independent clean references, spatially aligned with the processed crops, and identical in dimensions. This reference provenance must be verified before making a definitive PSNR/SSIM claim.

2. Problem Statement and Output Requirements

The objective is to recover readable license-plate text from images or video where the plate may be blurred, low-resolution, noisy, poorly illuminated, skewed, or captured at an angle.

The system is designed to return three principal outputs per input:

1. **Detected plate region:** bounding box and corresponding plate crop.
2. **Enhanced or corrected plate image:** rectified and/or adaptively enhanced crop.
3. **Recovered plate text:** normalized OCR candidate selected using confidence, syntax and consensus rules.

The implementation also includes a continuous-video mode with vehicle tracking, best-frame selection, appearance-based tracklet merging, and duplicate plate-read suppression.

3. System Architecture and Methodology

3.1 Single-image processing pipeline

Stage Method and purpose

1. Scene deblurring GoPro-trained NAFNet is applied to the full scene. Images with longest dimension above 1,024 px are downscaled; spatial dimensions are padded to multiples of 16. A detection-confidence fallback preserves the original scene if deblurring reduces plate-detection confidence.

2. Plate detection A dedicated YOLOv8 license-plate detector uses confidence threshold 0.20. The detected ROI receives an adaptive 8--10% margin to reduce clipping of edge characters.

3. Geometry correction Adaptive thresholding/Canny contour analysis and quadrilateral approximation are used to estimate plate corners. Ordered corners are rectified with a homography to a 520 × 130 (4:1) canvas.

4. Quality assessment Patch noise estimate, Laplacian/Sobel sharpness, normalized contrast spread and estimated character height guide processing decisions.

5. Adaptive enhancement An enhancement ladder retains raw and rectified candidates and may apply LAB-channel NLM denoising, CLAHE, unsharp masking, Real-ESRGAN x4 (native or downsampled 2x), and morphological residual re-inking. Candidates receive an 18 px white border before OCR.

6. OCR RapidOCR/PaddleOCR ONNX is run on standard and inverted grayscale versions of candidates to support different plate/background polarities.

7. Candidate selection OCR strings are normalized and compared using confidence, Indian registration syntax, position-aware character disambiguation and edit-distance/consensus scoring.

3.2 Adaptive quality routing

The quality-assessment stage is intended to avoid applying every expensive operation indiscriminately. The documented triggers are:

- Super-resolution when estimated character height is below 18 px or plate height below 35 px.
- Denoising when estimated noise sigma exceeds 4.5.
- CLAHE when normalized contrast spread is below 0.40.
- Sharpening when the sharpness index is below 45.

The routing engine therefore attempts to match processing intensity to the observed plate quality.

3.3 Video mode

For continuous footage, YOLOv8 and ByteTrack track cars, buses and trucks. A clarity score based on vehicle area, sharpness, aspect ratio and detector confidence selects a strong observation within each tracklet. A ResNet-18 appearance embedding (512-dimensional, L2-normalized) and cosine similarity threshold of 0.82 are used to merge likely broken tracklets within a 350-frame temporal window. Plate-level reads are subsequently deduplicated using text similarity, retaining the sharpest observation.

This video architecture is described as an implemented mode; the supplied 300-image evaluation does not quantify video tracking or Re-ID performance.

4. Dataset and Evaluation Protocol

The primary evaluation uses **300 prepared real-world image samples** with plate text and bounding-box annotations. The images were consolidated into a prepared dataset directory, while source provenance remains available in the manifest. No artificial degradation was added to this primary benchmark. The dataset was mostly clean or moderately degraded, which limits its ability to measure performance specifically on severe blur, strong sensor noise, or extreme low-resolution cases.

Three OCR configurations were compared:

1. Original ground-truth plate crop followed by OCR.
2. Enhanced ground-truth plate crop followed by OCR.
3. Full autonomous pipeline, including detection and subsequent processing.

The first two isolate OCR/enhancement on annotated plate crops; the third includes localization and geometry errors as well.

4.1 Metrics and justification

Metric	What it measures	Why it is relevant
Precision / Recall / F1	Correct plate detections at IoU 0.50	Measures whether the system finds the plate region.
AP@0.50 and mAP@0.50:0.95	Detection precision-recall quality; mAP averages across stricter IoU thresholds.	Detection performance at one threshold reflects localization tightness.
Mean IoU	Average overlap between predicted and ground-truth boxes; misses are counted as zero in the all-plates figure.	Measures bounding-box alignment, not just whether a plate was detected.
Exact-match accuracy	Fraction of plate strings that match the ground truth exactly.	A single wrong character makes a plate read incorrect; this is important for forensic use.
CER / character accuracy	Edit distance normalized by ground-truth character count; accuracy is 1 − CER under the report's definition.	Measures partial text recovery and character-level errors.
PSNR / SSIM	Pixel-level fidelity against a clean reference.	Potentially useful for restoration, but only with independent, spatially aligned clean reference images.
Latency / throughput	Mean processing time and frames per second.	Shows practical feasibility and identifies bottlenecks.

5. Quantitative Results

5.1 Detection and localization

Metric	Result
Precision @ IoU 0.50	99.66%
Recall @ IoU 0.50	99.00%
F1 @ IoU 0.50	0.9933
AP @ 0.50	0.9873
mAP @ 0.50:0.95	0.5336
Mean IoU (all plates; misses = 0)	0.7906
Mean IoU (detected only)	0.7959
Detection failure rate	0.67% (2 of 300)

The detector found almost all annotated plates at IoU 0.50. The gap between AP@0.50 (0.9873) and mAP@0.50:0.95 (0.5336) indicates that detections often overlap the target sufficiently at the 0.50 threshold, but box tightness is less consistent at stricter IoU thresholds.

Detection stratification by plate size

Plate size	Samples	Recall @ 0.50	Mean IoU (all)	Mean IoU bucket (detected)
Small (< 32^2 px)	34	100.0%	0.7061	0.7061

Medium 167 100.0% 0.7807 0.7807 (32^2--96^2 px)

Large (≥ 99 97.0% 0.8363 0.8536 96^2 px)

The small-plate group has the lowest mean IoU, suggesting less precise localization despite 100% recall at the stated threshold. The large group has the highest detected-only mean IoU, but two large plates were missed. These size buckets follow the evaluation code's area criteria.

Figure 1 --- YOLO license-plate localization IoU distribution

5.2 Geometric rectification

Metric	Measured value	Interpretation
Mean residual skew	0.23°	Low residual skew angle according to the implemented estimator; interpretation depends on how unmeasurable cases are handled.

Mean aspect-ratio 19.38% Above the stated error target of <15%; indicates that output geometry/aspect ratio remains an area for improvement.

The 0.23° residual-skew result should be interpreted cautiously if the estimator assigns zero when it cannot find usable lines. The available annotations contain axis-aligned bounding boxes rather than ground-truth quadrilateral corners, so they do not support a direct corner-error evaluation of perspective rectification.

5.3 OCR and end-to-end performance

Metric	End-to-end result
Samples	300
Exact-match accuracy	50.0%
Macro CER	0.1808
Macro character accuracy	81.92%
Corpus CER	0.1811
Corpus character accuracy	81.89%
Ground-truth characters	2,882
Total edit distance	522

The 50.0% exact-match rate means half of the evaluated plates were read with every character correct. Character-level accuracy is higher because many incorrect strings still contain several correctly recognized characters. For forensic use, exact-match performance remains especially important.

5.4 Three-way baseline comparison

Mode	Exact match	Mean CER	Macro char. accuracy	Corpus CER
Original GT crop → OCR	51.67%	0.1714	82.86%	0.1704
Enhanced GT crop → OCR	50.33%	0.1707	82.93%	0.1711
Full pipeline → OCR	49.33%	0.1845	81.55%	0.1842

On this predominantly clean/moderately degraded benchmark, enhancement did not improve aggregate exact-match accuracy: the enhanced ground-truth crop result is 1.34 percentage points below the raw-crop baseline. Mean CER is nearly unchanged, and macro character accuracy changes only slightly. The full pipeline

is lower than the raw ground-truth-crop baseline, which is expected to include additional detection and geometry errors as well as OCR errors.

This result does not establish that enhancement is ineffective on severely degraded plates. Since severe degradation was not well represented in the 300-image benchmark, aggregate scores may understate the benefit of enhancement on difficult inputs. During additional testing, the pipeline was observed to make some very poor-quality plates more readable and, in selected cases, help recover text that was not readily legible in the original. These observations are qualitative and should not be presented as a measured dataset-wide accuracy gain.

Figure 2 --- OCR baseline comparison

5.5 Supplementary PSNR / SSIM plate-only benchmark

The supplied supplementary CSV contains 18 finite paired records.

Metric Before After Mean change Paired outcomes

PSNR 13.09 dB 18.38 dB +5.28 dB 13 improved, 5 worsened SSIM 0.3695 0.7458 +0.3763 17 improved, 1 worsened

The mean values above are recalculated from those records. The paired changes are positive on average, with PSNR increasing in 13/18 records and SSIM increasing in 17/18 records.

Important metric-validity condition: PSNR and SSIM measure similarity to a reference, not enhancement quality in isolation. These values can be claimed as restoration-fidelity results only if the reference is an independent clean image of the same plate, with corresponding geometry, alignment and dimensions. If the reference is the raw input itself, the metrics instead measure similarity to that raw input and cannot establish recovery of the clean scene. The report therefore treats this 18-sample analysis as supplementary and conditional on confirming reference provenance.

The earlier seven infinite-PSNR records are not included in the supplied 18-row CSV; infinite PSNR usually indicates zero pixel error and should be investigated as a possible identical-image/self-reference case rather than silently converted to a finite score.

The primary 300-image dataset and this 18-sample image-quality subset are separate evaluations and must not be combined into one denominator.

5.6 Character-position accuracy

```
Character index   Recognition accuracy
```

```
              0                  74.0%
              1                  80.7%
              2                  82.7%
              3                  84.0%
```

```
4                77.0%
5                84.0%
6                87.0%
7                87.9%
8                88.3%
9                85.8%
```

The supplied positional curve shows weaker recognition at the first character position (74.0%) and at index 4 (77.0%), while positions 6--8 are stronger. This can help focus error analysis, but the positional values are aggregate character-position results and should be interpreted in the context of plate strings of varying lengths.

Figure 3 --- OCR recognition accuracy by character index

5.7 Character confusion analysis

Actual → predicted Count

M → H 11 0 → O 11 B → 8 10 4 → 2 9 1 → I 6 M → N 6

The alignment analysis also reports:

- **Top deletions:** M (32), H (28), 0 (27), 1 (26), 7 (21).
- **Top insertions:** 1 (8), E (4), N (3), K (2), F (2).

These patterns are consistent with character ambiguity, weak strokes, low contrast and crop/geometry errors.

5.8 Runtime and throughput

Pipeline stage Mean time (ms) Approx. share

Deblurring 576.21 15.5% Detection 66.24 1.8% Geometry 4.80 0.1% Quality assessment 10.11 0.3% Routing and restoration 204.21 5.5% OCR and selection 2,851.09 76.8% Total 3,713.00 100%

The measured throughput is 0.27 images/second (approximately 0.3 FPS). OCR and candidate selection dominate the runtime at about 76.8% of the total, followed by deblurring at about 15.5%. Therefore, optimizing detector inference alone is unlikely to materially improve end-to-end throughput. The evaluation metadata records device `cuda:0`; it does not establish that both available T4 GPUs were utilized.

7. Failure Cases and Limitations

7.1 Enhancement can be counterproductive on already readable plates

The 300-image comparison shows that enhancement does not universally improve OCR. On relatively clean images, denoising, sharpening, contrast equalization or super-resolution may alter stroke thickness, edges and character shapes without adding useful information. The enhanced GT-crop exact-match result (50.33%)

is below the raw GT-crop result (51.67%). The system's adaptive routing is intended to mitigate this, but the observed aggregate result shows that conservative processing and candidate selection remain important.

7.2 Severe blur and information loss

Motion blur and very low resolution are difficult because character strokes may merge or disappear. Deblurring and super-resolution can make patterns more legible, but cannot reliably reconstruct information that is absent. Generative restoration can produce plausible-looking details that are not faithful to the original plate; a visually cleaner character must not automatically be treated as a correct forensic recovery.

7.3 Noise and low illumination

Noise can resemble fine character strokes, while denoising can erase faint strokes. Low illumination and weak contrast reduce separation between glyphs and plate background. CLAHE and NLM may help selected cases, but aggressive settings can amplify artifacts or suppress character detail. The primary 300-image dataset contained relatively few severely noisy examples, so these degradation types were not adequately represented for a strong quantitative conclusion.

7.4 Geometry, clipping and border artifacts

Incorrect quadrilateral detection or corner ordering can warp character shapes. A fixed 4:1 output aspect ratio may not match every plate. Tight detector boxes can clip outer characters; margin padding and an OCR border are safeguards, but excessive padding or incorrect crops can also affect recognition. The measured mean aspect-ratio error of 19.38% exceeds the stated target of 15%, indicating room for improvement.

7.5 OCR character ambiguity and candidate selection

The most common substitutions include M/H, 0/O, B/8, 4/2 and 1/I. Position-aware disambiguation and Indian plate syntax can resolve some ambiguities, but syntax alone cannot prove a character identity. Incorrect syntax assumptions, non-standard formats, faint lettering or a cropped prefix/suffix can lead to a plausible but incorrect final string.

7.6 Runtime

OCR candidate generation and selection are the largest runtime bottleneck. Evaluating multiple enhancement candidates and both image polarities increases the number of OCR inferences. A future optimization could use a fast first-pass OCR and invoke expensive candidates only when confidence or syntax checks indicate uncertainty.

7.7 Evaluation limitations

- The 300-image benchmark is dominated by clean or moderately degraded images and is not a controlled severe-degradation benchmark.
- The 18-sample PSNR/SSIM subset is small and must be accompanied by explicit reference-image provenance and alignment details.
- Axis-aligned bounding boxes do not provide ground-truth plate corners, limiting direct rectification accuracy evaluation.
- The supplied results do not provide a quantitative evaluation of video tracking, Re-ID tracklet merging or temporal consensus.

- The reported single-device metadata is `cuda:0`; multi-GPU utilization should not be claimed without separate profiling evidence.

8. Discussion and Observations from Testing

The main observation is that detection is substantially stronger than exact text recognition. The system localizes almost every plate at IoU 0.50, but only half of the full-pipeline outputs match the complete ground-truth string. This suggests that the next accuracy improvements should focus on crop geometry, preservation of fine character strokes, OCR candidate efficiency and ambiguity resolution rather than only detector recall.

The baseline comparison also shows why enhancement must be evaluated by input difficulty. On the mostly clean primary dataset, enhancement yields no meaningful aggregate OCR gain. During additional testing on very poor-quality plates, however, the team observed cases where enhancement improved readability and enabled more useful OCR output. This is a plausible complementary strength, but it remains a qualitative observation because those severe cases were not collected as a separate, consistently annotated test set.

The system therefore uses an adaptive enhancement strategy rather than assuming every image benefits from every operation. Its candidate ladder retains raw and rectified alternatives alongside processed variants, allowing the OCR selection stage to choose among them. This is important in a forensic setting because aggressive enhancement can distort evidence; preserving the original crop and clearly distinguishing it from processed outputs is essential.

9. Conclusion

The project delivers an end-to-end forensic-oriented ANPR implementation with plate detection, geometric correction, adaptive restoration, OCR and a separate video tracking/deduplication architecture. On 300 real-world images, the detector achieved AP@0.50 of 0.9873 and recall of 99.0%, while end-to-end exact-match OCR accuracy reached 50.0% with corpus CER of 0.1811. Mean latency was 3.713 seconds per image.

The results indicate that plate localization is a relative strength, while exact OCR, geometric consistency and inference speed are the principal limitations. Enhancement did not improve aggregate OCR accuracy on the mostly clean evaluation set, but qualitative testing showed promising behaviour on selected severely degraded plates. A small supplementary PSNR/SSIM benchmark reports positive average changes, subject to verification that the reference images are independent, clean and spatially aligned.

Future work should prioritize a dedicated severe-degradation benchmark, stronger reference-based restoration evaluation, improved geometry validation, conservative handling of generative super-resolution outputs, OCR runtime reduction and expanded video-mode evaluation.

10. Deliverables and Reproducibility

Deliverable	Included / action
1. Source code	Organized source files and model assets; provide repository/package with the submission.

2. Technical report This report; update the five qualitative case paths and verify PSNR/SSIM reference provenance.

3. Five before/after examples Insert the five case outputs in Section 6 and include corresponding image files.

4. README Provide installation, model-weight setup, dataset/manifest format, command to run image mode, command to run video mode, and expected outputs.

End of Report